

# Lecture 5

## Discrete Random Variables

09.03.2026

# Outline

- 1 Discrete random variables
- 2 Expectation
- 3 Variance

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# Random Variables

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Example: If  $S = \{a = (a_1, a_2) : 1 \leq a_1, a_2 \leq 6\}$  is the 36 element space resulting from rolling two fair six-sided dice, then the following are all random variables

$$X(a) = a_1$$

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- A random variable is a number associated with a random experiment.
- Random variables are, in essence, a fancy way of describing an event, e.g.

$$P(X = 1) = 1/6$$

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$$P(X = 1) = P(\text{H}) = p$$

$$P(X = 2) = P(\text{TH}) = (1 - p)p$$

$$P(X = 3) = P(\text{TTH}) = (1 - p)^2 p$$

.....

$$P(X = n) = P(\text{TT} \cdots \text{TH}) = (1 - p)^{n-1} p$$

.....

# Cumulative distribution function (cdf)

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In the previous coin flipping example,

$$F(n) = P(X \leq n) = \sum_{i=1}^n (1-p)^{i-1} p = \frac{[1 - (1-p)^n]p}{1 - (1-p)} = 1 - (1-p)^n$$

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- ④  $F(x)$  is right continuous; i.e., for any decreasing sequence  $\{x_n : n = 1, 2, \dots\}$  that converges to  $x$ ,

$$\lim_{n \rightarrow \infty} F(x_n) = F(x) \implies \lim_{n \rightarrow \infty} P(X \leq x + \frac{1}{n}) = P(X \leq x)$$

The cdf tells us everything about a random variable  $X$

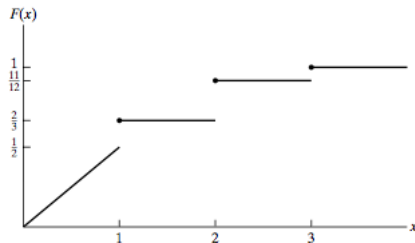


FIGURE 4.8: Graph of  $F(x)$ .

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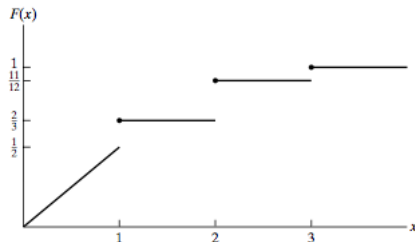


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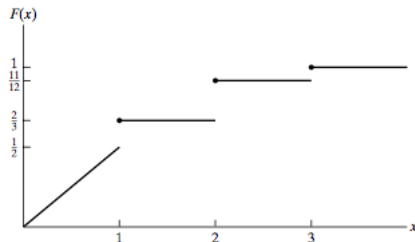


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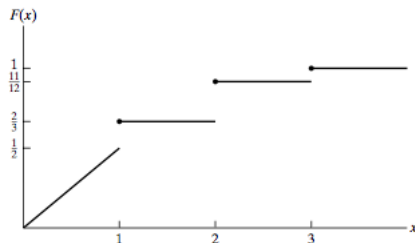


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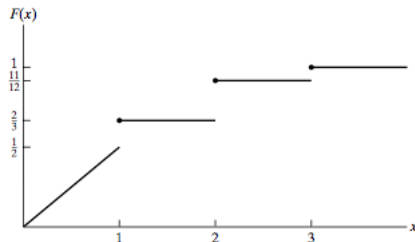


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- $X$  = number of your siblings.  
 $X$  can be  $0, 1, 2, \dots, \infty$ , countable.  $\implies$  discrete random variable.
- $X$  = a random number in  $[0, 1]$  generated by computer  
 $X$  can anything in  $[0, 1]$ , uncountable.  $\implies$  not discrete random variable.

# pmf and cdf

- For a discrete random variable  $X$ , there exists a countable sequence  $x_1, x_2, \dots$ , such that

$$p(x_i) > 0 \quad \text{for } i = 1, 2, \dots$$

$$p(x) = 0 \quad \text{for all other values of } x$$

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- Relationship between pmf and cdf (for discrete random variable)

$$F(a) = \sum_{\text{all } x \leq a} p(x)$$

- If we know pmf, we can compute cdf. And vice versa.

X has a pmf given by

$$p(1) = \frac{1}{4}, p(2) = \frac{1}{2}, p(3) = \frac{1}{8}, p(4) = \frac{1}{8}$$

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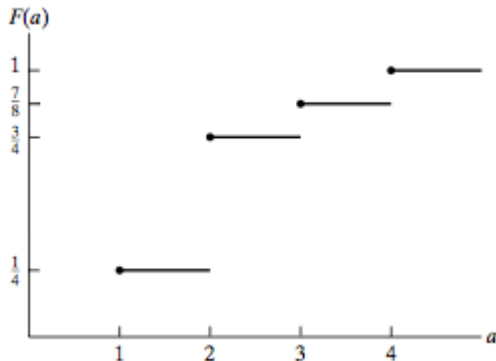
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$$F(a) = \begin{cases} 0 & a < 1 \\ \frac{1}{4} & 1 \leq a < 2 \\ \frac{3}{4} & 2 \leq a < 3 \\ \frac{7}{8} & 3 \leq a < 4 \\ 1 & 4 \leq a \end{cases}$$

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# Expected value

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- $E[X]$  is a weighted average of the possible values  $x$  that  $X$  can take on, each value being weighted by the probability  $p(x)$ .

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$$p(1) = p(2) = p(3) = p(4) = \frac{1}{4}$$

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# Expectation of a function of a random variable

- If  $X$  is a discrete random variable, and  $g$  is a real-valued function then the expectation (or expected value) of  $Y = g(X)$  is

$$E[g(X)] = \sum_{x:p_X(x)>0} g(x) \cdot p_X(x)$$

$X$  is a discrete random variable with pmf

|          |       |       |       |
|----------|-------|-------|-------|
| $x$      | $-1$  | $0$   | $1$   |
| $p_X(x)$ | $1/4$ | $1/2$ | $1/4$ |

$Y = X^2$ . Compute  $E[Y]$ .

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$$\begin{aligned} E[Y] &= \sum_{\text{all } x} x^2 \cdot p_X(x) \\ &= (-1)^2 \times \frac{1}{4} + 0^2 \times \frac{1}{2} + 1^2 \times \frac{1}{4} \\ &= \frac{1}{2} \end{aligned}$$

# Properties of expected values

If  $a$  and  $b$  are constants, then

$$E[aX + b] = aE[X] + b$$

- Holds for all random variable  $X$  (not necessary discrete random variable).

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- Proof for discrete random variable: let  $g(X) = aX + b$

Special cases of linear transformation  $E[aX + b] = aE[X] + b$

- constant factor

$$E[aX] = aE[X]$$

- constant

$$E[b] = b$$

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*Variance measures the variation (or spread) of these values*

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- One common simplification:

$$\text{Var}(X) = E(X^2) - \mu^2$$

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$$Var(X) = 7.5 - 2.5^2 = 1.25$$

$$SD(X) = \sqrt{Var[X]} = 1.12$$

# Variance measures the spread of $X$

$X_1$  is a discrete random variable with pmf

|              |       |       |
|--------------|-------|-------|
| $x$          | $-1$  | $1$   |
| $p_{X_1}(x)$ | $1/2$ | $1/2$ |

$X_2$  is a discrete random variable with pmf

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| $p_{X_2}(x)$ | $1/2$ | $1/2$ |

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$$\mu_1 = 0, \sigma_1^2 = E[X_1^2] - 0^2 = 1, \sigma_1 = 1$$

$$\mu_2 = 0, \sigma_2^2 = E[X_2^2] - 0^2 = 4, \sigma_2 = 2$$

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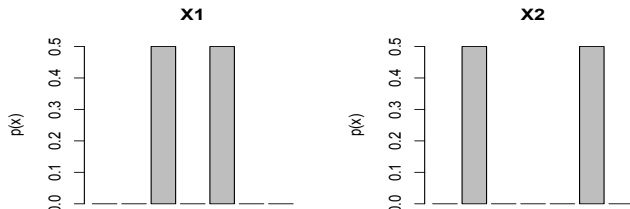
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$$\mu_1 = 0, \sigma_1^2 = E[X_1^2] - 0^2 = 1, \sigma_1 = 1$$

$$\mu_2 = 0, \sigma_2^2 = E[X_2^2] - 0^2 = 4, \sigma_2 = 2$$

Increasing variance (or sd) reflects increasing spread.



## Property of variance

We know that  $Var(X) = E[(X - \mu)^2]$ , and for constants  $a, b$ ,

$$E[aX + b] = aE[X] + b.$$

Write  $Var(aX + b)$  as a function of  $Var(X)$ .

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$$\begin{aligned} Var(aX + b) &= E[(aX + b - E[aX + b])^2] \\ &= E[(aX + b - aE[X] - b)^2] \\ &= E[(aX - a\mu)^2] \\ &= E[a^2(X - \mu)^2] \\ &= a^2 Var(X) \end{aligned}$$

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